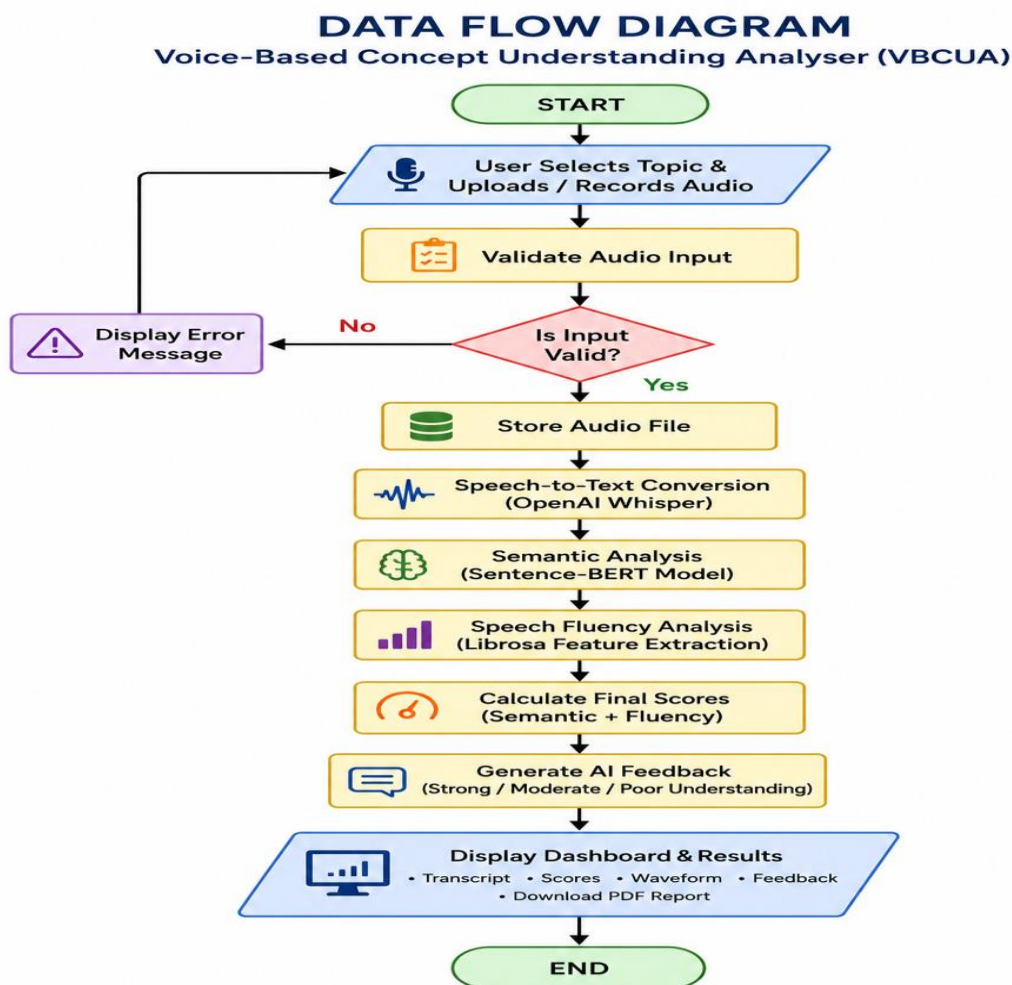


Data Flow Diagram

Date	10/07/2026
Name	Karuru Arjun
Project Name	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	2 Marks

Data Flow Diagram:



Flow Steps

1. Start
2. User Selects Topic and Uploads/Records Audio
 - The user opens the VBCUA application.
 - Selects a concept (e.g., Machine Learning).

- Records or uploads a voice explanation.
- 3. Validate Audio Input
 - The system checks whether the uploaded audio is valid and supported.
- 4. Decision: Is Input Valid?
 - No → Display an error message and ask the user to upload a valid audio file.
 - Yes → Continue to the analysis process.
- 5. Store Audio File
 - The uploaded audio is temporarily stored for processing.
- 6. Speech-to-Text Conversion
 - OpenAI Whisper converts the spoken explanation into text.
- 7. Semantic Similarity Analysis
 - Sentence-BERT compares the transcript with the reference concept.
 - A conceptual understanding score is generated.
- 8. Speech Fluency Analysis
 - Librosa extracts audio features such as:
 - Pause Ratio
 - RMS Energy
 - Speech Duration
 - Filler Words
- 9. Generate Final Evaluation
 - The system combines semantic and fluency scores.
 - AI generates qualitative feedback such as:
 - Strong Understanding
 - Moderate Understanding
 - Poor Understanding
- 10. Display Results
 - The dashboard displays:
 - Transcript
 - Semantic Similarity Score
 - Fluency Metrics
 - Waveform Visualization
 - AI Feedback
 - Overall Performance Score
- 11. Download PDF Report
 - The user downloads a structured PDF report containing all evaluation results.
- 12. End